UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the Interstate Transmission System	**Docket No. RM26-4-000**

COMMENTS OF INVENERGY WIND DEVELOPMENT NORTH AMERICA LLC, INVENERGY SOLAR DEVELOPMENT NORTH AMERICA LLC, AND INVENERGY THERMAL DEVELOPMENT HOLDINGS LLC

On October 23, 2025, the Secretary of Energy directed the Federal Energy Regulatory Commission ("FERC" or the "Commission") to initiate a rulemaking and consider the Department of Energy's ("DOE") Advance Notice of Proposed Rulemaking ("ANOPR"), pursuant to section 403 of the DOE Organization Act, to adopt rules to ensure the timely and orderly interconnection of large loads to the transmission system.[1] The ANOPR articulates fourteen principles intended to guide the Commission's effort to standardize the large load interconnection process. Invenergy Wind Development North America LLC, Invenergy Solar Development North America LLC, and Invenergy Thermal Development Holdings LLC (collectively, "Invenergy") hereby submit these comments in support of DOE's effort to establish new, clearly-defined rules to interconnect large loads while maintaining the Commission's bedrock principles of resource-neutrality, open access, and competition, and respecting state jurisdiction over retail rates, contracting terms, and permitting. Invenergy's comments focus on the substance of such rules and does not take a position on the Commission's jurisdiction over large load interconnection to the transmission system.

[1] Letter from Chris Wright, Secretary of Energy, Dep't of Energy, to Chairman Rosner and Comm'rs, Federal Energy Regulatory Commission (Oct. 23, 2025); Federal Energy Regulatory Commission, Notice Inviting Comments, Docket No. RM26-4-000 (Oct. 27, 2025).

I. COMMENTS

Utilities around the country are confronting an increasing quantity of large-load interconnection requests, largely due to accelerating demand for data centers and artificial intelligence ("AI"). In response, transmission owners across the country have adopted a hodge-podge of load interconnection procedures that often impose significant regulatory and financial burdens that threaten to stymie data center deployment and which to date have not served as an effective mechanism for screening non-speculative interconnection requests, reducing ratepayer costs, or increasing speed to power. In fact, the existence of disparate rules in each jurisdiction incentivizes large load customers to forum shop for the best rules by submitting interconnection requests in multiple service territories, thereby inducing speculative interconnection requests and increasing systemwide costs.

In an attempt to reduce such speculation, certain transmission owners have adopted unduly burdensome financial security and interconnection requirements that serve as an undue barrier to entry and reduce competition in what is otherwise a dynamic and growing industry. While financial security requirements should be sufficiently high to reduce speculation, they should not be arbitrary or untethered to a project's anticipated costs. And yet transmission owners too often require developers to post billions of dollars in security up front to interconnect hybrid facilities without any regard to the actual cost of interconnecting and serving data center load, in some instances requiring hybrid projects to securitize both the load *and* generation interconnection requests. Needless to say, these financial demands, which vary in amount by jurisdiction, can pose an undue barrier to the interconnection of new load and generation and are unjust and unreasonable to the extent they are not based on a project's anticipated costs.

In light of these constraints and to further our national AI priorities, Invenergy agrees with the ANOPR's proposal to standardize the large load interconnection process by adopting *pro forma* procedures that apply on an equal basis to all large load interconnection customers seeking access to the transmission system. Such procedures should prevent unduly disparate treatment, discourage forum shopping and speculative interconnection requests, and reduce systemwide costs by encouraging competition among large load interconnection customers under a common set of rules. Specifically, Invenergy proposes that any standardized large load interconnection process adhere to the following principles:

- Load and hybrid interconnection requests should be studied in the existing generator interconnection queue(s) with generator interconnection requests and not in a separate large load queue. Combining these processes will increase transparency for load and generation developers, reduce the need for costly network upgrades, and ensure equitable treatment of load and generation as they interconnect to the transmission system. Invenergy believes this approach has the greatest chance of fulfilling Principle Three in the ANOPR calling for studying load and generation together.
- Large load interconnection rules should be resource neutral, with no preference given to generation technology type. Market forces and competition are the best mechanisms to identify what generation resources can be timely and cost-effectively deployed to serve each type of large load. Large load customers should be free to select generation resources that meet their energy needs and corporate goals, provided they are consistent with state rules regarding who can supply large loads. A diverse array of generation resources will decrease systemwide costs, reduce supply chain constraints, improve reliability by reducing common-mode failure risk, and allow for quicker interconnection of load.
- The Commission should adopt commercial readiness criteria or security deposits for large load interconnection requests that correspond to potentially necessary network upgrades, taking into account the *net* impact of co-located load and generation together. Standardized rules that are consistent with the cost-causation principle will reduce speculative large load interconnection requests, unleash large load deployment by providing certainty for data center and generation developers, and remove undue barriers to entry for new market entrants. Right-sizing commercial readiness and security deposits will also allow developers to redeploy capital to build new generation and load, rather than unnecessarily tie up capital with security requirements that do not correspond with anticipated costs. Assessing net impacts from co-located loads and generators will avoid charging the load an

unduly high security deposit that raises costs to consumers and slows data center deployment without basis in potential network upgrade needs, which are smaller when co-located generation is taken into account.

A. Interconnection Process

1. Large Loads Should be Studied in a Combined Queue With Generation Interconnection Requests.

Invenergy agrees with DOE that load and hybrid facilities should be studied together with generation and encourages the Commission to require the study of load, hybrid facilities, and generation as part of a single interconnection queue. The current practice of studying load and generation interconnection requests in separate queues is an inefficient use of resources, increases ratepayer costs, and will create unnecessary delay as large loads and generators wait for their separate upgrades to be built. As DOE explained in the ANOPR, studying load and generation separately can, and often does, result in the identification of network upgrades that would not be needed had the facilities been studied together. Beyond this inefficiency, the more upgrades that are identified, the longer it takes to interconnect data centers so that they are fully functional and are not subject to curtailment or not allowed to energize until otherwise unneeded network upgrades are built.

Furthermore, because load interconnections, unlike generator interconnection requests, are studied exclusively by each transmission owner and typically only include transmission solutions in that particular transmission owner's footprint, the study process often overlooks potentially less-expensive transmission solutions that would have been identified by looking at the system more holistically, as it is in the generator interconnection study process. Unless large loads are allocated 100% of the upgrade costs to interconnect, which is not uniform practice among transmission organizations, non-large load ratepayers ultimately bear such costs of overbuilding or inefficiently building the transmission system. Furthermore, studying load and generator interconnection requests separately stifles generation development by depriving generation developers of useful insight into whether and when load may materialize.

DOE's proposal to study load and hybrid facilities together with generation would remedy these issues. A common interconnection queue could match the appropriate generation to load and thereby facilitate faster resource adequacy solutions to accommodate rapidly growing load, while also reducing the risk of constructing unnecessary network upgrades. Moreover, maintaining a single queue for generation, load, and hybrid facilities will increase transparency and limit opportunities for undue discrimination.

The Commission should also clarify that high voltage, direct current ("HVDC") solutions should be treated as equivalent to co-located generation in the integrated study process, including any expedited study process for co-located (or hybrid)[2] projects. Like generation, HVDC converter stations enable controllable injection of large quantities of electricity that can offset grid demand from large loads. HVDC converter stations also independently provide reactive power support that can obviate network upgrades that may otherwise be needed for large load interconnection, and like generation, they can also provide other ancillary services, such as voltage control, frequency control, black start, and ramping. Clarifying that HVDC converter stations should be eligible for expedited study as co-located projects with large loads would ensure a more comprehensive set of potential hybrid project arrangements that enable least-cost solutions for grid development, minimizing consumer impacts while accelerating data center deployment.

Finally, Invenergy recognizes that it may take time to fashion holistic interconnection rules to jointly study load and generation. As an interim measure to expedite the interconnection of large

[2] Invenergy understands the DOE Proposal's use of the term "hybrid" to mean a generator co-located with load seeking interconnection as one project.

loads and hybrid resources, the Commission could issue rules creating an expedited interconnection process for load that has contracted with a generator. This will provide a prompt avenue for those loads that are already ready to interconnect, while giving the Commission time to fashion a more comprehensive interconnection process that jointly studies generation and load.

Invenergy urges the Commission, while it undertakes these reforms, to adopt a transition process that does not prejudice existing large load interconnection customers that have already spent time and committed resources navigating existing state interconnection processes.

2. <u>The Commission Should Standardize Study Deposits, Readiness Requirements, and Withdrawal Penalties.</u>

Invenergy also agrees with DOE that load and hybrid interconnection requests should be subject to standardized study deposits, readiness requirements, and withdrawal penalties. Such requirements will help reduce speculative load interconnection requests, incentivize large loads to materialize and speed up the interconnection process. Requiring significant financial commitments and readiness requirements as part of a standard interconnection process can also have a positive effect on other aspects of the energy economy which depend on accurate load forecasts, such as the capacity market. By encouraging large load interconnection customers to only submit viable requests that they intend to pursue to completion it will reduce the incentive to forum shop for the best rules in multiple jurisdictions, right size load forecasts and eliminate phantom load.

In addition to adopting standardized financial security requirements, Invenergy recommends that the Commission require transmission providers to produce transparent, publicly accessible, and timely load forecast information that distinguishes between confirmed and speculative load. Such information would incentivize generation development by providing more reliable data on where new generation resources should be located in order to best serve confirmed load.

While Invenergy agrees with DOE's proposal to adopt standardized interconnection rules, it is critical that such rules not act as an undue barrier to entry for viable load and hybrid projects. Large load interconnection customers are currently subject to a wide variety of security requirements. In some jurisdictions Invenergy has observed exorbitant security requirements for load paired with merchant generation that effectively force load to partner with load serving entities in order to secure preferable terms.[3] The unjust nature of these financial demands is exacerbated by the fact that such security is often not based on the actual cost to construct the network upgrades required to accommodate the load interconnection. As a result, there is no incentive for transmission owners to choose the most cost effective or timely transmission solution. Rather than protect against the interconnection customer's non-performance (the ostensible purpose of security payments), these types of arrangements stymie development and create undue barriers to market entry.

Because the current patchwork of interconnection rules promotes anticompetitive behavior and discourages even viable load interconnection requests, Invenergy recommends the Commission adopt security and study deposit requirements that are consistent with Commission's cost causation policy and generator interconnection rules. As with generation interconnection requests, study deposits for large load and hybrid facilities should be based on the size of the request. For hybrid interconnection requests, there should be only one study deposit that is based on the greater of the generation size or load size, but not both. Any commercial readiness or security required should be determined based on the amount of the network upgrades required to interconnect the load, consistent with the generator interconnection rules and the Commission's

[3] For example, Tri-State recently proposed $2.7 million/MW for large load security, which FERC stated was not shown to be appropriately sized to mitigate the identified risks. *Tri-State Generation and Transmission Association, Inc.*, 193 FERC ¶ 61,070, at P 54 (2025).

cost causation policy.[4] Such a standardized policy will prevent undue discrimination against individual market participants while ensuring that the interconnection rules for load and hybrid interconnection do not stifle load or generation development.

In addition to standardized study deposits and financial readiness requirements, the Commission should also consider site control requirements for load and hybrid interconnections to incentivize viable requests. For hybrid interconnection requests, the generation component should conform to the current site control requirements approved in Order No. 2023, which is 90% generation site control at the time of application based on resource acreage.[5] For the load component or load seeking interconnection without generation, 100% site control of the facility site should be required at application. As for site control for interconnection facilities, Order No. 2023 did not establish site control requirements.[6] But several regional transmission organizations and transmission owners have implemented such requirements.[7] Invenergy recommends that any interconnection facility site control requirements for load or hybrid interconnection be the same as those required for generation interconnection as reflected in the existing tariffs.

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[4] *See, e.g., Reform of Generator Interconnection Procs. & Agreements*, Order No. 845-A, 166 FERC ¶ 61,137, at P 78 (2019) ("[I]t would be inconsistent with the cost causation principle to exempt an interconnection customer from interconnection facility and network upgrade costs that would not be necessary but for that interconnection request."); *Standardization of Generator Interconnection Agreements & Procs.*, Order No. 2003, 104 FERC ¶ 61,103, at P 694 (2003) ("it is appropriate for the Interconnection Customer to pay initially the full cost of Interconnection Facilities and Network Upgrades that would not be needed but for the interconnection."); *see also K N Energy, Inc. v. FERC*, 968 F.2d 1295, 1300 (D.C. Cir. 1992) ("Simply put, [the cost-causation principle] has been traditionally required that all approved rates reflect to some degree the costs actually caused by the customer who must pay them.").

[5] *Improvements to Generator Interconnection Procs. & Agreements*, Order No. 2023, 184 FERC ¶ 61,054, at P 594, *order on reh'g*, 185 FERC ¶ 61,063 (2023), *order on reh'g*, Order No. 2023-A, 186 FERC ¶ 61,199, at P 199 (2024) (maintaining the 90% site control requirement).

[6] Order No. 2023, 184 FERC ¶ 61,054, at P 604; Order No. 2023-A, 186 FERC ¶ 61,199, at P 197.

[7] *See, e.g.,* MISO Tariff, Attachment X, Generator Interconnection Procedures, §§ 7.2.1, 7.2.1.1 (outlining site control requirements for interconnection facilities).

Finally, currently load serving entities often require detailed specifications for data center projects at the time of application including ramp rate schedules, planned total investment, number of jobs created, water usage, and local ordinance approvals, among other detailed information about planned facilities. This information is not often available to generation and data center developers at the outset of the interconnection process and bears no relevance to studying the interconnection or whether a rule is "just and reasonable" or "not unduly discriminatory" under the Federal Power Act. Nor do these requirements exist for the interconnection of generation or transmission. The Commission should not propose or permit transmission owners to propose technical requirements that are immaterial to a project's interconnection and which create unnecessary barriers to entry or that preference load serving entities or transmission owners. The Commission has held that site control and commercial readiness deposits are the best proxy to ensure that generation facilities are ready to interconnect. It should hold that the same is true for load and hybrid interconnections.

3. Load and Hybrid Facilities Should be Allocated 100% of the Network Upgrade Costs They Cause.

Invenergy agrees with DOE's proposal to make load and hybrid facilities responsible for 100% of the network upgrades that they are assigned through any interconnection studies, rather than allocate such costs to non-large load customers or generators in the interconnection queue. This will ensure that costs are assigned in accordance with the Commission's cost causation principle, which requires entities to bear the costs that they cause, and will further mitigate the likelihood of widespread withdrawals of generation from the interconnection queue.

With regard to studying a hybrid facility for network upgrade cost allocation, as a default, hybrid projects should be studied as one project for purposes of assigning network upgrade costs, rather than studying the load and generation separately. However, if the hybrid project chooses, it should be given the option to be studied as load only or generation only and each component allocated the network upgrades costs they individually cause. This would allow hybrid projects to understand the full suite of interconnection possibilities and potential cost allocation in the event the load or generation terminates before the other.

II. COMMUNICATIONS

Please address all notices and communications regarding this proceeding to the following persons who are also designated for service in this proceeding:

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III. CONCLUSION

For the foregoing reasons, Invenergy respectfully requests that the Commission standardize the large load interconnection process by adopting *pro forma* procedures consistent with the recommendations herein.

Respectfully submitted,

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CERTIFICATE OF SERVICE

I hereby certify that on this 21st of November, a copy of the foregoing document has been electronically served upon each person designated on the official service list in this proceeding.

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